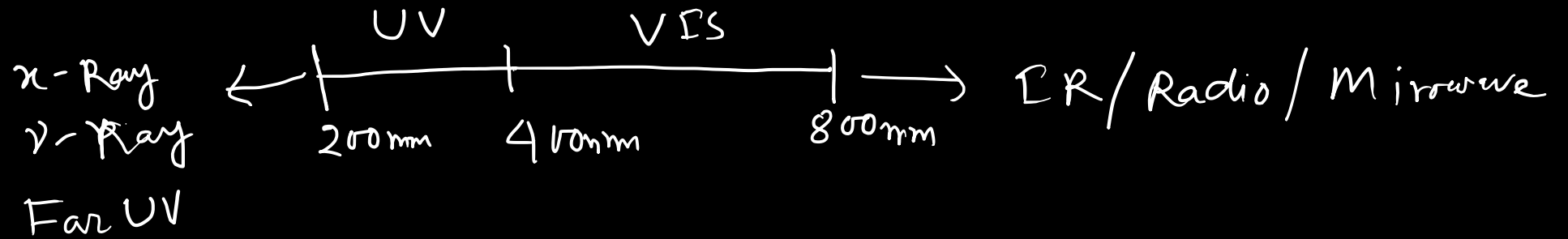
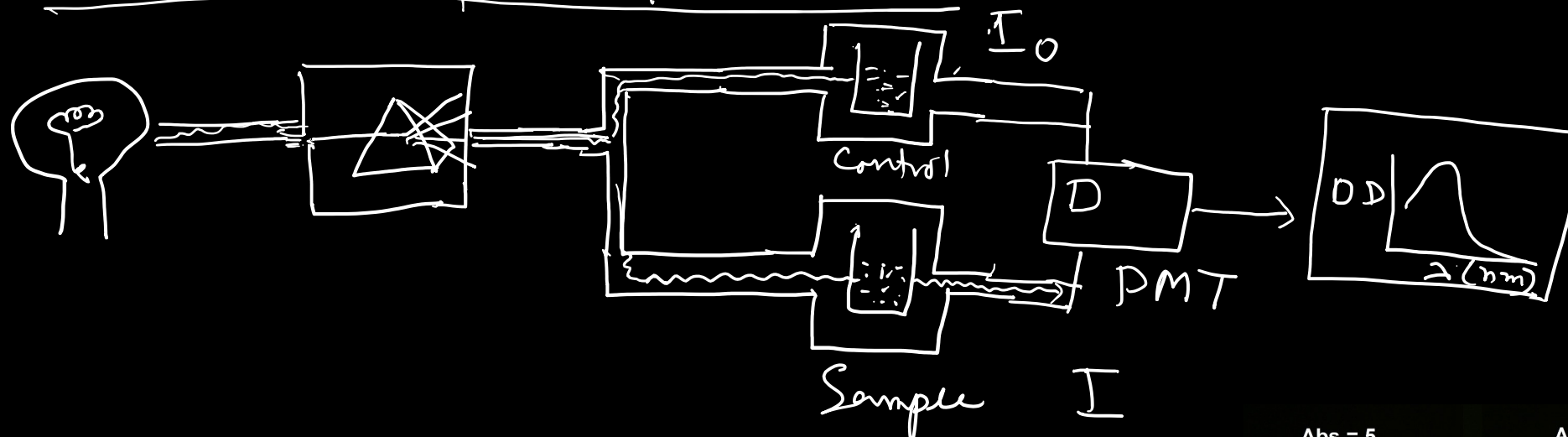


UV- VIS Spectroscopy



- Application:
- ① To know the exact con^n of a solⁿ (Main)
 - ② To know the presence of Chemical Gr.
Any molecule can absorb light in the range of UV- VIS if it has a chromophore.
- Chemical Group.

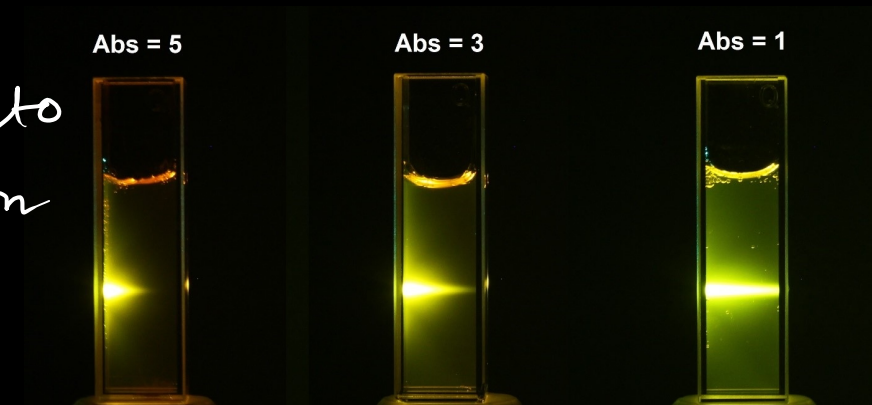
Set-Up: Spectrophotometer



Loss of light intensity due to Absorption of Sample is known as Absorbance.

$$A = \frac{I_0}{I}$$

• Transmittance (T) = $\frac{1}{A} = \frac{I}{I_0}$



Case I
Transparent
 $I = I_0$
 $T = 100\%$

Case II
 $I = 0$
 $T = 0\%$

$$\text{Absorbance } (A) = \log_{10} \left(\frac{1}{T} \right)$$

① Lambert's Law: Absorbance of light $\left[\log_{10} \left(\frac{I_0}{I} \right) \right]$ in homogeneous solⁿ is directly proportional to the length of sample in which the light passes

$$\text{So, } A = \log \left(\frac{I_0}{I} \right) \propto l$$

② Beer's Law: Absorbance of light in sample is directly proportional to the conⁿ of solⁿ in which light passes

$$A \propto C$$

Combining these two laws: Lambert-Beer's Law —

$$A \propto c \times l$$

$$= \epsilon \times c \times l$$

↑
molar extinction coefficient

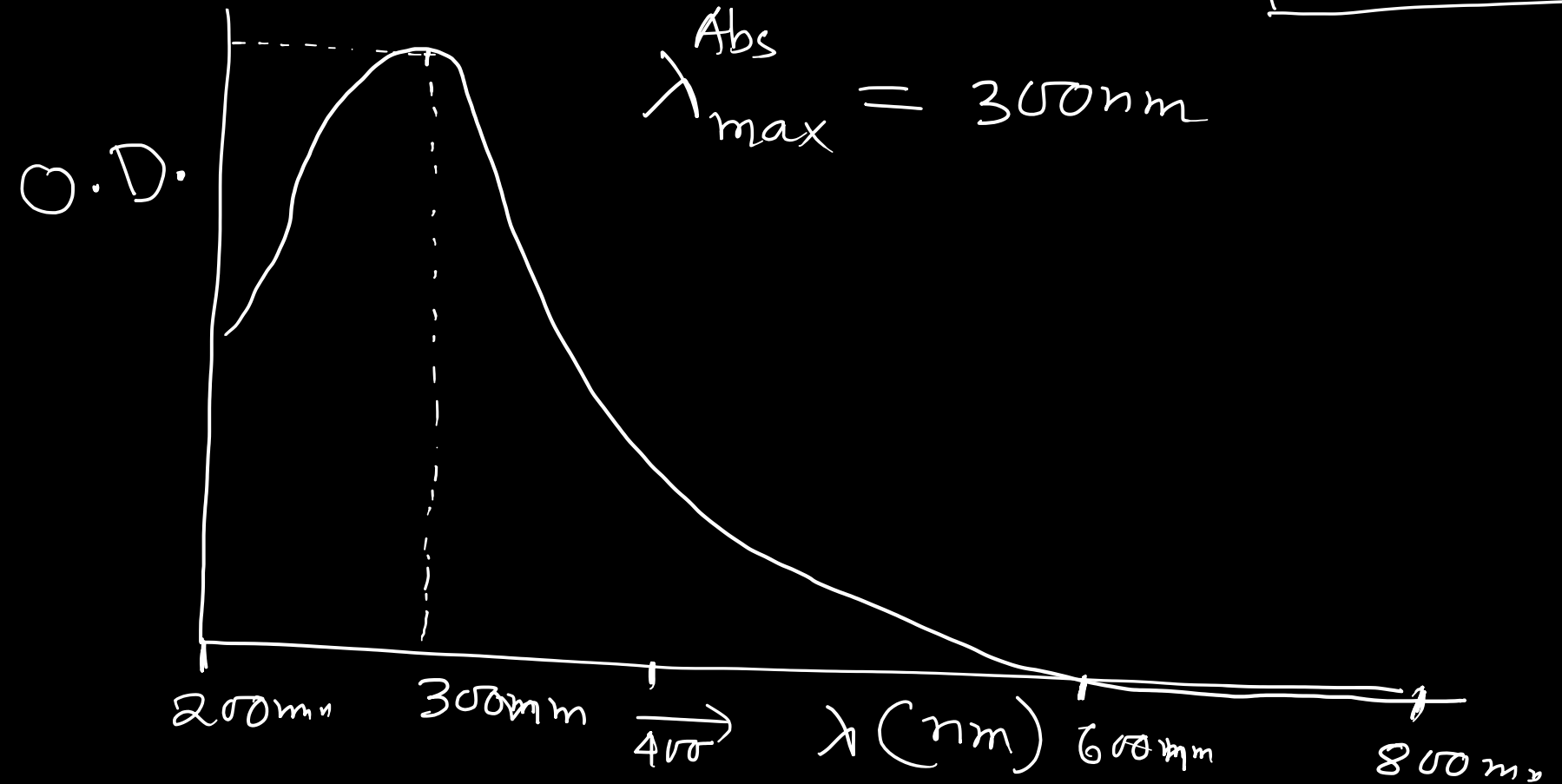
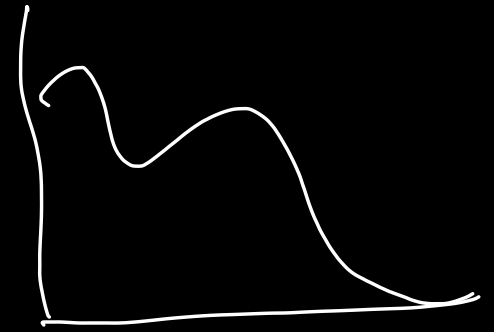
Unit of ϵ :

$$\text{Unitless} = ? \times \frac{\text{mol}}{\text{L}} \times \text{cm}$$

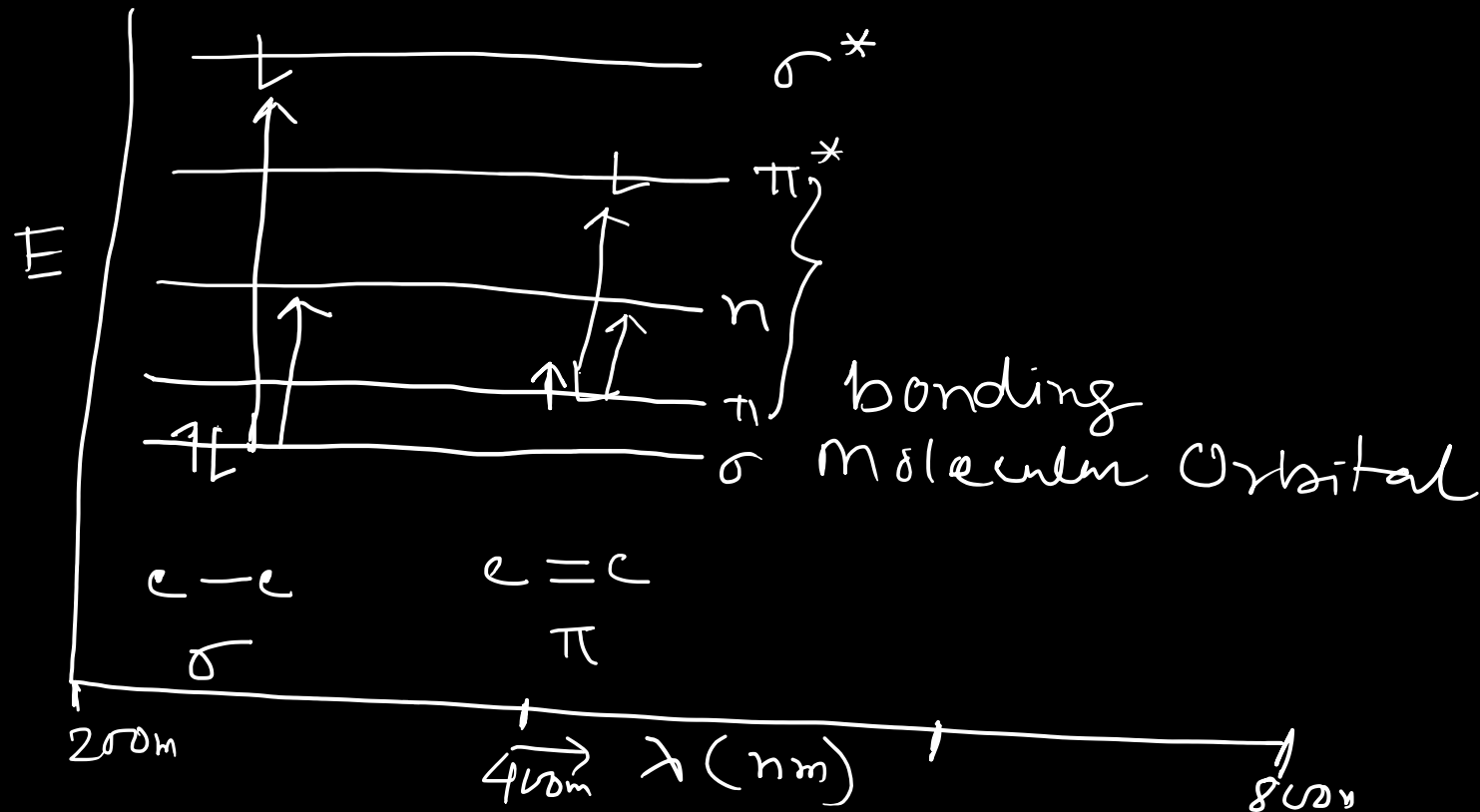
$$\boxed{\frac{\text{L}}{\text{mol} \times \text{cm}}}$$

$$\text{L} / \text{mol} \cdot \text{cm} \text{ or } \text{L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$$
$$\text{m}^{-1} \cdot \text{cm}^{-1}$$

Data from Spectrophotometer



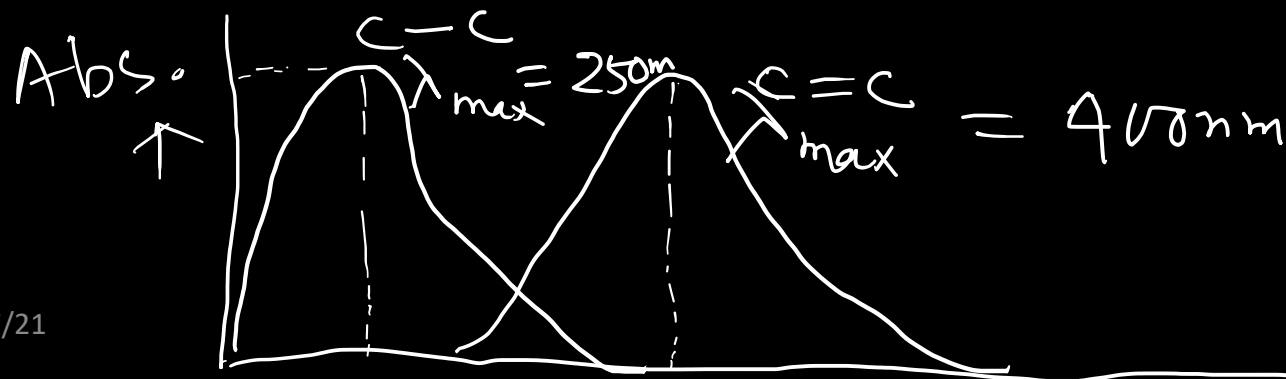
Mol. mechanism of Absorption



$$E = h\nu$$

$$= \frac{hc}{\lambda}$$

$$E \propto \frac{1}{\lambda}$$



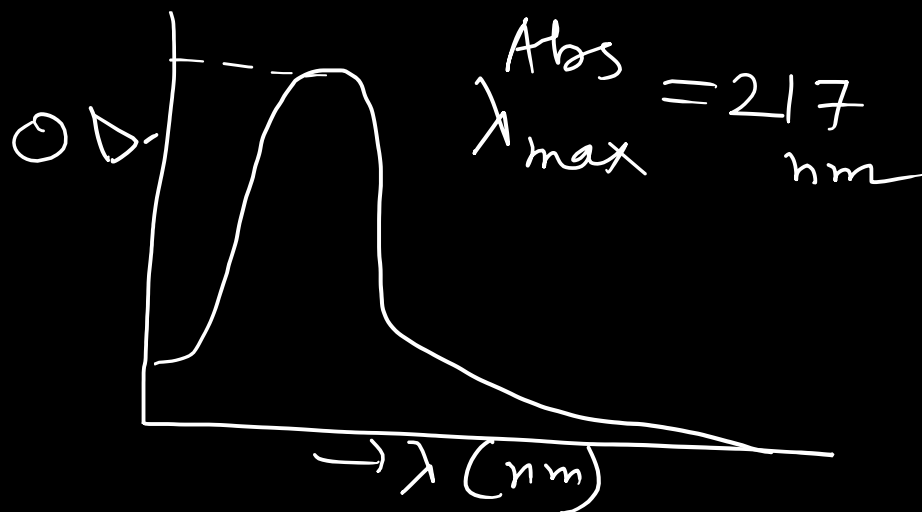
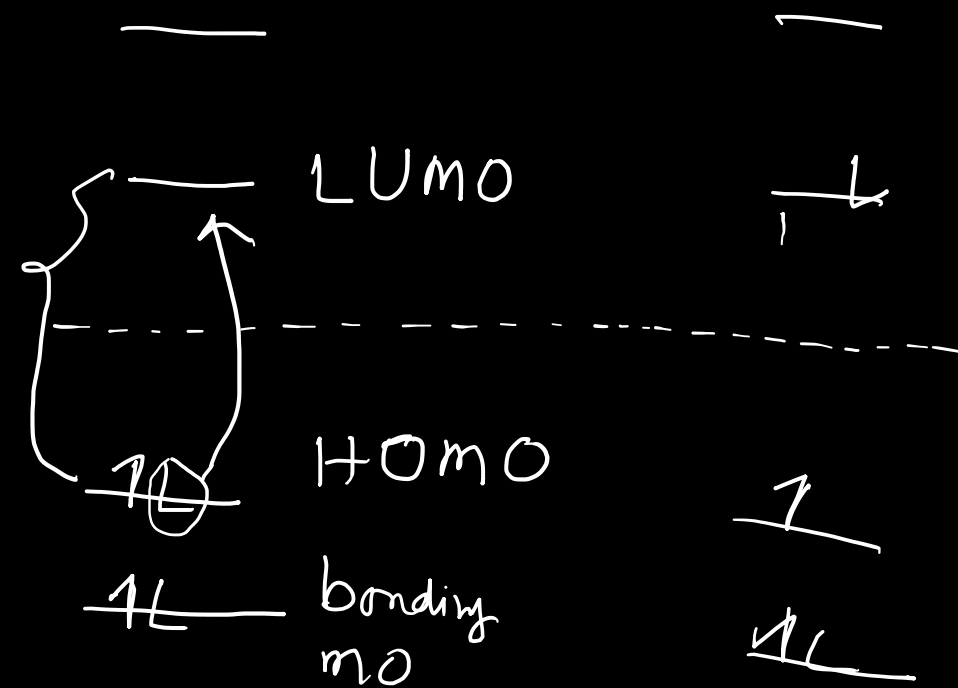
Specific Example:

$4\pi e^-$



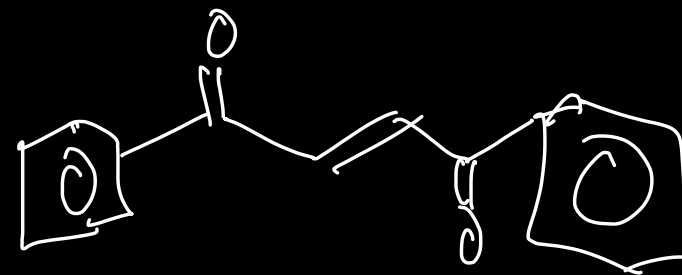
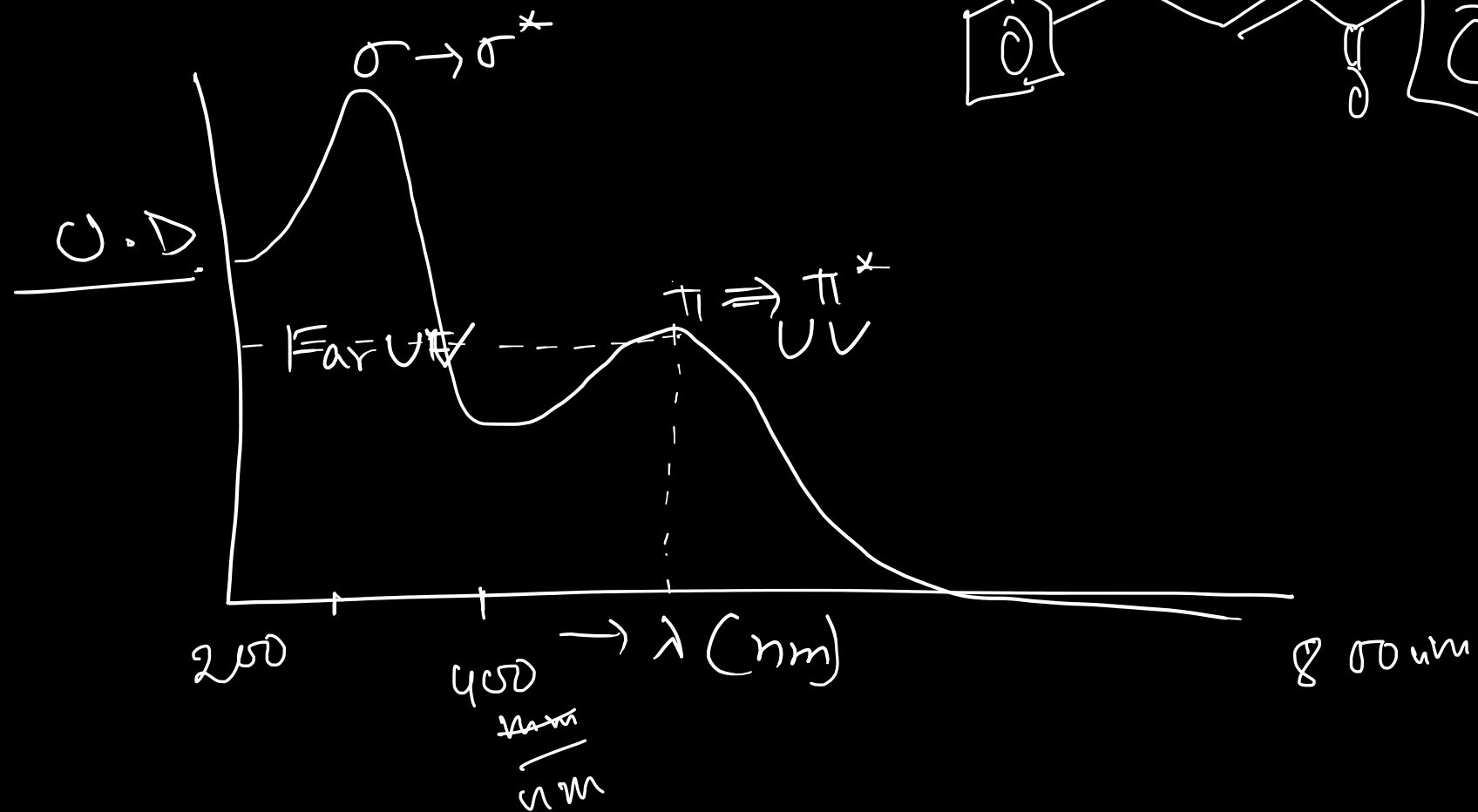
1,3-butadiene

4 A.O. $\rightarrow$ 4 MO.



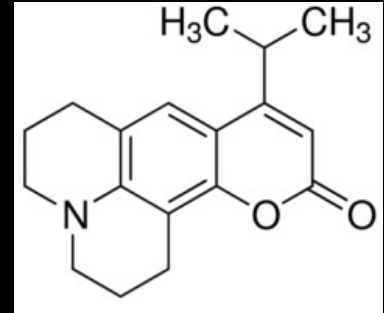
Ground State

Excited State



Q1: How to determine the conⁿ of a dye solⁿ (C 480)

- ① prepare a solⁿ in H₂O
- ② put inside cuvette
- ③ measure background
- ④ measure Abs.



$$OD = \epsilon \times c \times l$$

$$\epsilon = \frac{OD}{c \times l}$$

$$= \frac{0.5}{85,000 \times 1 \text{ cm}} \rightarrow \text{Reported}$$

$$= \frac{0.5}{85,000 \times 1 \text{ cm}} \rightarrow 60,000 \rightarrow 200,000$$

